

Exact Structural Abstraction and Tractability Limits

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Abstract

Any computational specification can be written as a state-indexed admissible-output relation. Exact relevance certification asks which coordinates are necessary to preserve correctness with respect to such a specification under coordinate deletion, and Section 6 proves that decision, counting, search, approximation guarantees, PAC/regret/risk guarantees, finite-horizon or anytime guarantees, and distributional specifications all reduce to this same coordinate-dependence problem. The tractable families treated here admit a finite primitive basis, but optimizer-quotient realizability is maximal, so quotient shape alone cannot characterize the frontier.

We prove a meta-impossibility theorem for efficiently checkable structural predicates invariant under the theorem-forced closure laws of exact certification. Structural convergence with zero-distortion summaries, quotient entropy bounds, and support-counting arguments explains why those closure laws are canonical. We establish the theorem by constructing same-orbit disagreements for four obstruction families, namely dominant-pair concentration, margin masking, ghost-action concentration, and additive/statewise offset concentration, using action-independent, pair-targeted affine witnesses. Consequently no correct tractability classifier on a closure-closed domain yields an exact characterization over these families. More generally, restricting to a closure-closed subdomain helps only if it removes all orbit gaps. Here closure-orbit agreement is forced by correctness, not assumed as an invariance axiom. The same compute-cost statement holds for polynomial-time optimizer computation. The obstruction therefore appears at the level of computational specifications themselves, not at the level of a particular output formalism.

1 Introduction

A computational specification is determined by the outputs that count as correct at each state, equivalently by a state-indexed admissible-output relation. Exact relevance certification asks which coordinates are necessary to preserve correctness with respect to such a specification under coordinate deletion. Under succinct encodings the ambient problem is hard, coNP-complete in static settings and PSPACE-complete in sequential settings [29], so any tractability statement must be genuinely structural. Section 6 proves three facts. First, every specification expressible as a state-indexed admissible-output relation reduces to exact relevance certification once written in that form; this includes decision, counting, search, approximation guarantees, PAC/regret/risk guarantees, finite-horizon or anytime guarantees, and distributional specifications with a fixed distribution (Proposition 6.23 and Corollaries 6.24–6.29). Second, semantic claims depend only on admissible-output equivalence (Proposition 6.33). Third, every state equivalence relation is realizable as an output specification of this kind (Corollary 6.34). The optimizer-quotient framework of [29] therefore isolates the canonical semantic layer on which feature sufficiency is determined. Section 6

also shows independently that exact certification itself factors through this same object: exact certification, sufficiency, certification statistics, and zero-distortion summaries all factor through the optimizer quotient (Proposition 6.1, Corollary 6.2, Corollary 6.4, and Proposition 6.5). The categorical route [29] and the operational route of Section 6 therefore identify the same object. This paper asks whether that canonical specification-level object admits a finite structural tractability classifier at the representation level.

The orbit-gap contradiction depends only on closure-law invariance (Theorem 4.9), and closure-law invariance is itself forced by correctness (Theorem 6.17). Closure-equivalent representations encode the same certification problem, so a correct tractability classifier must assign them the same verdict. The same forced-agreement statement extends to polynomial-time solvability of optimizer computation, with canonical instantiations for deterministic payload output and admissible-output search built from optimizer sets, so the no-go applies uniformly to the compute-cost layer as well as to feature sufficiency. Restricting the domain is therefore not a loophole in kind: Corollary 4.14 shows that it helps only if the restricted closure-closed domain eliminates all orbit gaps for the target notion. The specific admissibility package of Definition 6.16 is one natural instantiation, but the package itself is not load-bearing.

On the positive side, we prove that the tractable families treated here split into three roles: primitive core mechanisms, regime lifts, and degenerate collapses. This yields a finite primitive basis. On its own, however, this does not establish a frontier theorem: finite inventory is not finite characterization.

The natural next move is to classify by optimizer-induced quotient structure. We show this is impossible in the unconstrained sense: optimizer realizability is maximal, and arbitrary labeling kernels are realizable. Thus quotient shape is too expressive to carry the boundary by itself.

Exactness is not arbitrary. At zero distortion, any summary that preserves optimal actions must still separate distinct optimizer classes: if two different optimizer classes were assigned the same summary value, the summary would merge states with different optimal-action sets and would cease to be lossless. Exact relevance therefore marks the theoretical floor for lossless abstraction. Section 6 now also fixes the approximation boundary: approximate surrogate claims about relevance or sufficiency are admissible only with an explicit stability reduction to the exact optimizer sets, because uniform error below a witness-gap bound preserves the corresponding witness while arbitrarily small uniform perturbations can otherwise flip the exact judgment.

The realizability barrier motivates the organizing principle of the paper: tractability predicates must be evaluated on representation-level structural classes, but modulo theorem-forced equivalences of presentation. In particular, closure under relabelings, positive affine utility reparameterizations, duplication, and binary irrelevant-coordinate extension is not a modeling preference; it is inherited from the fact that exact certification depends only on the induced decision quotient relation.

Standing Semantics

For a decision problem with state space S , action space A , and utility U , write

$$\text{Opt}(s) = \{a \in A : U(a, s) \text{ is maximal among all actions at } s\}.$$

This induces the decision-equivalence relation

$$s \sim_{\text{Opt}} s' \iff \text{Opt}(s) = \text{Opt}(s').$$

The associated optimizer quotient is the quotient set $S/\sim_{\text{Opt}}$. A coordinate set I is *sufficient* when agreement on I forces equality of optimizer classes, and a coordinate is *relevant* when deleting it

destroys sufficiency. These are the semantic objects studied throughout the paper; later closure laws and obstruction families are all phrased relative to this quotient viewpoint.

At the same time, unrestricted predicates trivialize any impossibility theorem via oracle encoding. We therefore isolate an admissibility layer: polynomial-time checkability, structural extractability, closure-law invariance, and bounded-pattern definability. This class excludes direct semantic encoding while still covering the structural results established here. The proof-load-bearing clause is closure-law invariance (Theorem 4.9); the remaining clauses act as guardrails that keep the search space structural rather than oracle-like.

This is a no-go for classifiers inside the admissibility class, not a no-go for tractable classes themselves. Stronger representation-sensitive structure can lie outside Definition 6.16; for example, the algebraic invariants used in CSP dichotomy theorems are not bounded-pattern definable in the present sense. This is why CSP-style positive frontier theorems are compatible with the present result rather than in tension with it. The open problem is therefore not to evade the theorem, but to identify a successor admissibility class that uses stronger structure of that kind.

Within this admissibility class, we prove a four-family no-go template. For each obstruction family we construct two slices with different obstruction status in the same closure orbit. The key mechanism is an action-independent, pair-targeted additive affine term (the statewise “ α -component” of positive affine transport), which changes the obstruction predicate while preserving closure equivalence. Closure-law invariance then transports any candidate predicate across the orbit and forces contradiction.

The dominant-pair family is the canonical instance, and the same mechanism now applies to margin masking, ghost-action concentration, and additive/statewise offset concentration. Theorem 4.9 identifies closure-law invariance as the only load-bearing hypothesis, and Theorem 6.17 upgrades the scope from invariant classifiers to correct classifiers on closure-closed domains. The remaining admissibility clauses serve as guardrails. The open problem is to identify a stronger structural principle, if one exists, that yields a correct frontier theorem.

An accompanying Lean development mechanically verifies the realizability constructions, the generic orbit-gap argument, and the family-level admissible no-go theorems used in the collapse argument; the archived artifact is available at <https://doi.org/10.5281/zenodo.19457896>.

2 Finite Basis for the Current Positive Landscape

We distinguish between *tractable families treated here* and *primitive tractability mechanisms*. A tractable family is any named positive condition included in the present analysis. A primitive mechanism is a smaller conceptual source of tractability that may explain several families at once. The positive theory organizes into three roles: six core structural families, four regime lifts, and five degenerate collapses. Projecting those families to primitive mechanisms leaves eight sources of tractability: the six core subcases together with constant-optimizer collapse and finite explicit enumeration.

Theorem 2.1¹ (Explicit Partition of the Current Positive Landscape). *The fifteen tractable families considered here admit an explicit disjoint partition into three blocks:*

1. *six core structural families,*
2. *four lifted families that reduce to existing core subcases, and*
3. *five degenerate families that become tractable by optimizer collapse or finite explicit enumeration.*

¹ Lean: FR1-2

Moreover, each family is classified into exactly one of these three blocks by an explicit classification map.

Proposition 2.2² (Explicit Inventory). *The fifteen tractable families considered here are exactly the following:*

1. *core: bounded actions, separable utility, low tensor rank, tree structure, bounded treewidth, coordinate symmetry;*
2. *lifts: product distribution, bounded support, bounded horizon, full observability;*
3. *degenerate: single action, bounded state space, strict global dominance, constant optimal set, multiplicative-separable constant-sign.*

Corollary 2.3³ (Finite Basis for the Current Positive Landscape). *Projecting the partition of Theorem 2.1 to primitive mechanisms yields an explicit finite list of eight primitive tractability mechanisms. The list consists of the six core structural subcases*

bounded actions, separable utility, low tensor rank, tree structure, bounded treewidth, coordinate symmetry, together with two degenerate mechanisms:

constant-optimizer collapse and finite explicit enumeration.

Proposition 2.4⁴ (Explicit Primitive Basis). *The current primitive basis consists exactly of bounded actions, separable utility, low tensor rank, tree structure, bounded treewidth, coordinate symmetry, constant-optimizer collapse, and finite explicit enumeration.*

The current positive landscape therefore contains fifteen tractable families accounted for by eight primitive mechanisms.

Proposition 2.5⁵ (All Independent Core Mechanisms Are Nondegenerate). *The analysis already contains witness families showing that five core mechanisms are individually indispensable in the sense of basis coverage:*

1. *a low-rank-only witness family,*
2. *a separable-only witness family,*
3. *a bounded-actions-only witness family,*
4. *a bounded-treewidth-only witness family, and*
5. *a symmetry-only witness family.*

Each of these families remains tractable by its designated mechanism while failing the corresponding versions of the other independent core mechanisms.

Proposition 2.5 settles the independent part of the basis question on the present surface. Every core mechanism that is not merely contained in another is shown to be genuinely needed.

Concretely, the present theory contains five kinds of “only-by” witness: low-rank only, separable only, bounded-actions only, bounded-treewidth only, and symmetry only.

² : FR25–28 ³ : FR3 ⁴ : FR29 ⁵ : FR101, FR105–106, FR108, FR112–113

Proposition 2.6⁶ (Weak Tree Ordering Does Not Imply Width One). *There exists a dependency presentation satisfying the older monotone tree-order predicate **TreeStructured** whose dependency graph has real treewidth greater than one. Width-one containment requires the stricter parent-tree condition from Proposition 2.8, not merely monotone ordering.*

Proposition 2.7⁷ (Cyclic Dependencies Recover Hardness). *The reduction family for exact relevance certification yields **coNP**-hard instances with cyclic dependency structure. Thus the tree-structured regime is not only sufficient for tractability; leaving it also reaches the hard side.*

Proposition 2.8⁸ (Parent-Tree Structure Gives Width-One Decompositions). *Suppose a dependency presentation is tree-structured in the strict parent sense: every dependency points to a smaller coordinate and every coordinate has at most one parent. Then the induced dependency graph admits a genuine tree decomposition of width at most one.*

Proposition 2.8 is the containment theorem behind the intended “tree structure is the treewidth-one special case” slogan. It distinguishes this parent-tree notion from the weaker monotone-order predicate currently named **TreeStructured**.

Proposition 2.9⁹ (Current Positive Interfaces Factor Through Role Classification). *The current positive theorem surface factors through the role classifier of Theorem 2.1. Concretely, the family-indexed interfaces for*

1. *subcase-facing tractability,*
2. *degenerate-case witnesses, and*
3. *complexity-class summaries*

all factor through the map

$$\text{family} \longmapsto \text{role} \in \{\text{core}, \text{lifted}, \text{degenerate}\}.$$

Proposition 2.9 shows that the exported positive API is already controlled by the role partition rather than by fifteen unrelated theorem schemas.

3 Lifts and Degeneracies

The extra named families beyond the six core subcases split naturally into two groups.

Regime Lifts

Four stochastic/sequential positive results do not introduce new primitive mechanisms. They reduce directly to existing core subcases:

1. product distributions reduce to separable utility;
2. bounded support reduces to bounded actions;
3. bounded horizons reduce to bounded treewidth;
4. full observability reduces to tree structure.

⁶ : FR114 ⁷ : FR115 ⁸ : FR100, FR110–111 ⁹ : FR4

These are mathematically useful because they transfer the static tractable theory into richer regimes. But they do not enlarge the primitive basis. They are best understood as *lifts*, not new frontier points.

Proposition 3.1¹⁰ (Lifted Cases Introduce No New Primitive). *Each stochastic/sequential tractable case considered here is classified into one of the six core structural subcases. In particular, these positive results do not enlarge the primitive basis from Corollary 2.3.*

Degenerate Cases

The remaining positive cases considered here are degenerate in a precise sense.

Single-action systems, strict global dominance, constant optimal-set systems, and multiplicative-separable constant-sign models all collapse the optimizer so strongly that the relevant-coordinate question disappears: the optimal-action set is constant across states, so the empty coordinate set is sufficient. These are not new structural tractability mechanisms for the relevance problem; they are constant-optimizer collapses.

Bounded state space behaves differently. It does not force a constant optimizer. Instead, it makes exact certification feasible by brute-force enumeration over a finite universe. This is polynomial only because the instance family already carries a hard external bound on its search space.

Proposition 3.2¹¹ (Degenerate Positive Cases). *Every non-core tractable case considered here that is not a regime lift belongs to one of two degenerate buckets:*

1. *constant-optimizer collapse, or*
2. *finite explicit enumeration.*

Taken together with Proposition 3.1, Proposition 3.2 sharpens Theorem 2.1: every positive result considered here is exactly one of three things, a primitive core case, a lift, or a degenerate collapse.

4 Toward the Frontier Theorem

Sections 2 and 3 motivate a frontier question. Do the tractable families already suggest a finite optimizer-compatible criterion? In the strongest form, could exact relevance certification admit a representation-sensitive dichotomy theorem analogous to the classical dichotomy programs for satisfiability and constraint satisfaction?

The most direct version of that program fails. Quotient shape is too expressive, and later in the section even structurally restricted finite classifiers fail on a theorem-forced surface. Any future frontier theorem must therefore avoid both sources of failure.

Three obstacles remain. Realizability is not the bottleneck: Theorem 5.1 shows that arbitrary labeling kernels already arise as optimizer quotients, so no frontier theorem can be driven by quotient realizability alone. Even on the positive side one must separate primitive mechanisms from lifts and degeneracies; otherwise a finite inventory confuses genuine sources of tractability with derived or collapsed cases. And although the long-term target remains an algebraic description analogous to the CSP dichotomy program, the negative results below show that no direct optimizer-compatible classifier can play that role.

¹⁰ : FR5 ¹¹ : FR1, FR6

A Stabilized Binary Pairwise Subregime

One piece of the frontier program already admits a clean canonical theorem. On binary coordinate domains, define the mixed difference along coordinates i, j for action a by evaluating the utility on the four assignments $(0, 0), (1, 1), (0, 1), (1, 0)$ while holding all other coordinates fixed at 0. For pairwise utilities, this mixed difference is the canonical witness of genuine pair interaction.

Proposition 4.1¹² (Binary Pairwise Symmetry Dichotomy). *Let the coordinate alphabet be binary. If a utility admits a pairwise decomposition and is invariant under coordinate permutations, then either:*

1. *it admits a unary-coordinate decomposition, so every pairwise term collapses into single-coordinate contributions, or*
2. *every distinct coordinate pair carries a genuine pair interaction, witnessed by a nonzero mixed difference for some action.*

Within the binary pairwise-symmetric regime, there is no intermediate sparse interaction pattern hiding between unary collapse and the complete-pair case. Once a single nonzero pair witness exists, symmetry propagates it to every coordinate pair. This removes one plausible source of additional primitive mechanisms from the frontier search.

The optimizer-relevant version needs one extra layer. Raw pair interaction is still too coarse, because action-independent pairwise terms can be structurally dense without changing the optimizer. The correct invariant is the mixed difference of action gaps $U(a, \cdot) - U(b, \cdot)$.

Proposition 4.2¹³ (Decision-Relevant Binary Pairwise Dichotomy). *In the same binary pairwise-symmetric regime, either all action-dependent pair effects collapse into a unary-coordinate reduction after removing an action-independent base state term, or every distinct coordinate pair is decision-relevantly interacting for some action gap.*

A coarse hardness heuristic therefore fails. The obstruction theorem shows that even complete genuine pair interaction and failure of coordinate symmetry do not force hardness by themselves: one can still have a constant optimizer. Any eventual frontier theorem must therefore quotient out optimizer-degenerate phenomena, not just declared or utility-level interaction density.¹⁴

The obstruction is stronger than that first formulation suggests. Action-specific utility offsets leave the decision-relevant interaction graph unchanged, yet they can turn a family with nonconstant optimizer behavior into one with a constant optimizer while preserving dense decision-relevant structure. Any true hardness theorem in this regime must therefore normalize away action offsets explicitly, not merely require dense decision-relevant interaction.¹⁵

Identify utilities up to action-dependent, state-independent offsets, and the offset-normalized decision-relevant graph becomes well-defined on those equivalence classes. But the resulting dichotomy attempt still fails. The offset-normalized obstruction theorem shows that even after passing to offset classes, one can retain complete decision-relevant interaction and unbounded treewidth while exact certification remains trivial.¹⁶

The next refinement is to restrict the interaction graph to optimizer-supported actions. This removes two earlier collapse mechanisms: the offset-collapse family and a ghost-action family whose all-action decision-relevant graph is complete even though a single coordinate remains sufficient. But even that support-filtered graph is not enough. The optimizer-supported obstruction theorem exhibits a margin-masking family with complete optimizer-supported decision-relevant interaction and unbounded treewidth, yet the optimizer still depends only on coordinate 0. The third collapse

¹² : FR118–120 ¹³ : FR121–124 ¹⁴ : FR124 ¹⁵ : FR125–127 ¹⁶ : FR128–129

mechanism is therefore not ghost support or offset collapse, but large unary margins that mask dense supported pairwise interactions.¹⁷

One might hope that a strict unary-to-pair margin bound rescues the dichotomy. Define *margin-bounded* pairwise utilities by requiring every unary term to be at most twice the largest binary mixed-difference magnitude. This still fails. The dominant-pair family is margin-bounded because its unary terms vanish, while its optimizer-supported decision-relevant graph remains complete. Yet the obstruction theorem shows that exact certification is still controlled by the two-coordinate set $\{0, 1\}$. The fourth collapse mechanism is therefore pair-weight concentration rather than unary masking: one supported pair dominates the optimizer while the remaining dense interactions are too weak to matter.¹⁸

Admissible Closure-Orbit No-Go Program

The obstruction families above lead to a uniform negative statement because they admit explicit disagreements inside a single closure orbit.

Proposition 4.3¹⁹ (Orbit-Gap Template). *Let Q be a slice predicate. Suppose there exist slices U, V such that U and V lie in the same closure orbit, $Q(U)$ holds, and $Q(V)$ fails. Then no closure-law-invariant predicate P can satisfy*

$$P(W) \iff Q(W) \quad \text{for all slices } W.$$

Proof idea. Closure-law invariance gives $P(U) \iff P(V)$. Exact agreement with Q would then imply $Q(U) \iff Q(V)$, contradicting the assumed orbit gap. ■

For each obstruction family, the argument constructs two same-orbit slices with different obstruction status. In all four cases the witness is a positive-affine step whose α -component is an action-independent state term supported on a single coordinate pair. Pure relabeling cannot change the relevant obstruction statistics, and a global scale factor preserves all pairwise magnitudes up to common rescaling. The statewise additive term is what allows one to move mass onto a chosen pair while remaining inside the theorem-forced closure laws. The same mechanism works for four structurally different obstruction families, and Proposition 4.12 together with Corollary 4.13 shows that orbit gaps are the complete obstruction criterion in the closure-invariant and correctness-forced regimes.

There is also a transport interpretation of these witnesses. Once transport is computed on optimizer-quotient classes, a singleton quotient admits zero-cost diagonal transport, whereas genuine branching forces positive off-diagonal transport when distinct classes carry mass. The pair-targeted affine witnesses exploit exactly this sensitivity. They do not introduce arbitrary perturbations unrelated to the decision geometry; rather, they move mass across a selected branch of the quotient while staying inside the same closure orbit. The affine witnesses therefore track the intrinsic transport geometry of the quotient instead of functioning as purely syntactic tricks.²⁰

Theorem 4.4²¹ (Dominant-Pair Admissible No-Go). *No admissible normalization predicate can decide the dominant-pair obstruction family exactly.*

Proof sketch. The target predicate is unique dominant-pair status: one pair/action realizes the maximal interaction magnitude and that maximizer is unique. Start from a slice whose unique dominant pair is $\{0, 1\}$. Add an action-independent affine state term supported on a different pair, say $\{1, 2\}$. This produces a slice in the same closure orbit but with a different unique dominant

¹⁷ : FR130–136 ¹⁸ : FR137–140 ¹⁹ : FR185 ²⁰ : FR183–184 ²¹ : FR186

pair. Proposition 4.3 then rules out every closure-law-invariant classifier, hence every admissible one. ■

Theorem 4.4 gives the first instance. The remaining three families use the same orbit-gap scheme, but each isolates a different structural obstruction.

At the compute-cost layer, the same dominant-pair orbit witness yields a machine-checked no-go for optimizer computation: no correct classifier for polynomial-time solvability of optimizer computation can decide dominant-pair status exactly. In particular, no admissible predicate can simultaneously track optimizer-computation polynomiality and exact dominant-pair status.²²

Theorem 4.5²³ (Margin-Masking Admissible No-Go). *No admissible normalization predicate can decide margin-boundedness exactly.*

Proof sketch. For margin masking, the target predicate is margin-boundedness itself. This family is structurally different from the dominant-pair family because it is governed by a threshold comparing unary mass against the largest pair interaction. The affine witness leaves unary terms unchanged but raises the largest pair interaction past the relevant threshold, so the translated slice becomes margin-bounded while the base slice is not. Proposition 4.3 therefore rules out every closure-law-invariant classifier, hence every admissible one. ■

Theorem 4.6²⁴ (Ghost-Action Admissible No-Go). *No admissible normalization predicate can decide the ghost-action concentration signature exactly.*

Proof sketch. For ghost actions, the target predicate says that there is an action whose unary contribution on the first coordinate is -1 on both binary values and whose anchor-pair interaction has unit mixed-difference magnitude. This family is local in appearance but unstable under the same pair-targeted affine move. A pair-supported affine term preserves the closure orbit while destroying the signature, so Proposition 4.3 applies. ■

Theorem 4.7²⁵ (Offset Admissible No-Go). *No admissible normalization predicate can decide the additive/statewise offset signature exactly.*

Proof sketch. For additive/statewise offset concentration, the target predicate requires two actions with distinct anchor-pair interaction magnitudes, namely 1 and 0. This family isolates the action-specific mismatch that survives after earlier offset normalizations. A pair-supported affine term changes the anchor-pair statistics while remaining in the same closure orbit. Again, Proposition 4.3 yields the contradiction. ■

Theorem 4.8²⁶ (Admissible Collapse Across the Four Obstruction Families). *Let P be a normalization predicate satisfying the admissibility axioms. Then P is not a tractability characterization.*

Proof idea. If P were an admissible tractability characterization, it would have to decide each of the four obstruction predicates above. Theorem 4.4, Theorem 4.5, Theorem 4.6, and Theorem 4.7 show that this is impossible. Hence no admissible normalization predicate yields the desired frontier theorem. ■

The four obstruction families are witnesses for the orbit-gap template (Proposition 4.3), not the scope of Theorem 4.8. The theorem ranges over the entire admissibility class. Four structurally distinct families are included because the same pair-targeted affine transport defeats classifiers across families that share no surface structure, and Proposition 4.12 shows that orbit gaps are the complete

²² : FR250–251 ²³ : FR187 ²⁴ : FR188 ²⁵ : FR189 ²⁶ : FR190

obstruction criterion for exact classification by closure-law-invariant predicates. The four families are therefore theorem-backed witnesses of a complete mechanism rather than an ad hoc list.

Abstractly, let Γ be a class of slice predicates, and call a predicate Γ -admissible when it belongs to Γ . Call Γ *closure-sound* if every Γ -admissible predicate is closure-law invariant. Call Γ a *reasonable guardrail package* if it is closure-sound and also imposes auxiliary restrictions such as efficient checkability, structural extractability, and bounded-pattern definability. The proof below uses only closure-soundness; the remaining clauses explain which non-oracular search spaces the no-go is meant to cover.²⁷

Theorem 4.9²⁸ (Closure-Sound Package No-Go). *Let Γ be any class of slice predicates such that every Γ -admissible predicate is closure-law invariant. Then no Γ -admissible predicate can simultaneously characterize the four obstruction families. In particular, the no-go theorem depends only on closure-law invariance; the remaining clauses of Definition 6.16 justify the admissibility package as a meaningful notion but do not enter the contradiction.*

Proof sketch. Closure-law invariance suffices to invoke Proposition 4.3 against each of the four obstruction families. The same orbit-gap proofs underlying Theorem 4.4, Theorem 4.5, Theorem 4.6, and Theorem 4.7 then yield the contradiction directly. Any one of the four families independently witnesses the collapse; we state all four to make explicit that the obstruction is robust across structurally distinct witnesses. ■

Theorem 4.9 is the load-bearing form of Theorem 4.8. Any reasonable formalization of admissibility, whether the specific package of Definition 6.16 or any alternative whose admissible predicates are closure-law invariant, inherits the impossibility conclusion. Proposition 6.1 identifies closure-law invariance as the semantic core, and Theorem 6.17 shows that correctness forces the same invariance on any tractability classifier. The admissibility package is one natural instantiation; the theorem is robust to its replacement.

Corollary 4.10²⁹ (Reasonable Guardrail Packages). *The same conclusion holds for every reasonable guardrail package. In particular, efficient checkability, structural extractability, bounded-pattern definability, and the informal exclusion of oracle encodings are guardrails rather than proof-load-bearing hypotheses.*

Proof. Immediate from Theorem 4.9, since only the closure-soundness clause is used. ■

Corollary 4.11³⁰ (No Correct Tractability Classifier). *Let C be any correct tractability classifier on a domain that is closed under the closure laws and contains the four obstruction families. Then C cannot yield an exact tractability characterization across those families. In particular, no procedure assigning verdicts to representations can correctly predict tractability across the four obstruction families, regardless of whether the procedure tracks closure-invariant features internally.*

Proof. By Theorem 6.17, any such classifier must agree on closure orbits and is therefore closure-law invariant on its domain. Theorem 4.9 then rules out exact characterization across the four obstruction families. ■

Corollary 4.11 is the witness-level universal form of the no-go. Theorem 4.9 isolates closure-law invariance as the load-bearing hypothesis, and Theorem 6.17 removes even that as an explicit assumption by forcing the same invariance from correctness itself. Corollary 4.14 then states the general domain-relative form: restricting the domain helps only if it removes all orbit gaps for the target notion.

²⁷ : FR191–192 ²⁸ : FR193 ²⁹ : FR194 ³⁰ : FR193, FR197–199

Proposition 4.12³¹ (Orbit-Gap Completeness for Exact Classification). *For any slice predicate Q , the following are equivalent:*

1. *there exists a closure-law-invariant predicate P such that $P(S) \iff Q(S)$ for every slice S ;*
2. *Q is itself closure-law invariant;*
3. *Q is constant on closure orbits.*

Equivalently, exact classification of Q by closure-law-invariant predicates fails if and only if Q admits an orbit-gap witness: two closure-equivalent slices with different Q -status.

Proof sketch. The implication $(1) \Rightarrow (2)$ transfers closure-law invariance across exact equivalence. The implication $(2) \Rightarrow (3)$ says exactly that a closure-law-invariant predicate cannot change value inside a closure orbit. For $(3) \Rightarrow (1)$, every primitive closure-law step is already a closure equivalence, so orbit constancy supplies the six invariance clauses. The final equivalence is the contrapositive of orbit constancy. ■

This is the precise true version of mechanism completeness used in the paper. The four obstruction families are not claimed to exhaust all hard-side phenomena. Rather, orbit gaps are the complete obstruction criterion for exact classification of any fixed target predicate by closure-law-invariant classifiers, and the four families provide structurally distinct witnesses of that criterion.

Corollary 4.13³² (Orbit-Gap Completeness on Closure-Closed Domains). *Let D be a closure-closed domain of slices, and let T be a target predicate on D . Then T admits an exact characterization on D by a closure-law-invariant predicate if and only if T has no orbit-gap witness inside D . Equivalently, exact characterization on D by closure-law-invariant predicates fails if and only if there exist $U, V \in D$ in the same closure orbit with different T -status.*

Proof sketch. This is the domain-restricted version formalized in Lean. The forward implication is immediate from closure-law invariance. For the reverse implication, take the closure hull of $D \cap T$; if T has no orbit-gap witness inside D , that closure hull agrees with T on D . ■

Applied to exact tractability, Theorem 6.17 places every correct tractability classifier on a closure-closed domain inside this regime. Orbit gaps are therefore the complete obstruction criterion for the universal-scope no-go once correctness is imposed.

Corollary 4.14³³ (Domain Restriction Helps Only by Removing Orbit Gaps). *Let D be a closure-closed domain, and let Q be a target predicate on D such that correctness of a classifier for Q forces closure-orbit agreement on D . Then D admits a correct classifier for Q if and only if Q has no orbit-gap witness inside D . Equivalently, restricting the domain avoids the no-go only by eliminating all orbit gaps of Q on that restricted domain.*

Proof sketch. The reverse implication is Corollary 4.13. For the forward implication, correctness-forced orbit agreement rules out any same-orbit disagreement inside D . The artifact proves this abstract principle directly, and also formalizes the optimizer-computation instance as a concrete compute-cost theorem. ■

This is the precise answer to the domain-restriction objection. Excluding the four named witness families is not enough by itself. Those families witness orbit gaps, but the logical boundary is orbit-gap freedom on the restricted closure-closed domain. If another orbit gap remains, the same impossibility theorem applies; if no orbit gap remains, the closure-hull construction yields a correct classifier on that domain.

³¹ : FR201-204 ³² : FR210-213 ³³ : FR252-256

5 The Realizability Barrier

One possible route to a finite tractability taxonomy would be to show that optimizer-induced quotients realize only a narrow subclass of equivalence relations, so that the frontier would emerge from realizability alone. In the unconstrained setting, this route fails.

Theorem 5.1³⁴ (Every Labeling Kernel Is Optimizer-Realizable). *Let $\phi : S \rightarrow T$ be any function. There exists a decision problem with action space T and state space S such that for every state $s \in S$,*

$$\text{Opt}(s) = \{\phi(s)\}.$$

Consequently, for all states $s, s' \in S$,

$$\text{Opt}(s) = \text{Opt}(s') \iff \phi(s) = \phi(s').$$

Equivalently, the decision quotient of the constructed problem is exactly the kernel partition of ϕ .

Proof. Take the action set to be T itself and define

$$U(a, s) = \begin{cases} 1 & \text{if } a = \phi(s), \\ 0 & \text{otherwise.} \end{cases}$$

Then the unique maximizer at state s is the designated action $\phi(s)$, so the optimizer set is exactly $\{\phi(s)\}$. Two states therefore have the same optimizer set if and only if they receive the same label under ϕ . ■

Theorem 5.1 is the key negative fact for the frontier program. It shows that optimizer realizability by itself is extremely permissive. If arbitrary label kernels are already optimizer-induced, then no finite taxonomy can come merely from asking which abstract quotients are realizable.

Proposition 5.2³⁵ (Every Equivalence Relation Is Optimizer-Realizable). *For every equivalence relation on the state space, there exists a decision problem whose decision quotient relation is exactly that equivalence relation. Moreover, the quotient object of the realizing problem is canonically equivalent to the corresponding setoid quotient.*

Proof. Let $\approx$ be an equivalence relation on S , and let $\pi : S \rightarrow S/\approx$ be the quotient map. Apply Theorem 5.1 to the labeling map π . The resulting decision problem satisfies

$$\text{Opt}(s) = \text{Opt}(s') \iff \pi(s) = \pi(s'),$$

and the right-hand side is exactly the original equivalence relation. Since the realizing quotient identifies states precisely by equality of quotient labels, its quotient object is canonically equivalent to the setoid quotient itself. ■

Proposition 5.2 is the maximal realizability statement available in the present framework. The labeling-kernel theorem is a special case. Thus the realizability barrier is not merely broad; at the level of abstract quotient relations, it is maximal.

Proposition 5.3³⁶ (The Realized Quotient Is Exactly the Label Range). *For every labeling map $\phi : S \rightarrow T$, the decision quotient of the realizing problem is canonically equivalent to the range of ϕ . In finite settings, the quotient cardinality is therefore exactly $|\text{range}(\phi)|$.*

³⁴ : FR7-8 ³⁵ : FR45-47 ³⁶ : FR39-40

Proof. By Theorem 5.1, two states lie in the same decision class if and only if they receive the same label under ϕ . Therefore the quotient class of a state depends only on the value $\phi(s)$, and sending the class of s to $\phi(s)$ defines a well-defined map from the decision quotient to $\text{range}(\phi)$. This map is surjective by definition of the range and injective because equality in the range means equality of labels, hence equality of quotient classes. ■

Proposition 5.3 strengthens Theorem 5.1. The realizing construction does not merely realize the induced equivalence relation abstractly; it realizes the quotient object itself with exactly the expected image size.

Corollary 5.4³⁷ (Realizability Alone Cannot Be the Frontier). *Any future optimizer-compatible tractability metatheorem must use more than quotient realizability alone. Additional restrictions have to enter through representation, coordinate structure, utility structure, closure properties of the family under study, or comparable algebraic invariants.*

This shifts the burden of the metatheorem. The hard question is not which quotients can arise in principle; almost all kernels can. The hard question is which *natural structural families of decision problems* force tractability or preserve hardness once coordinate structure and utility representation are taken seriously.

6 Necessary Conditions on Any Frontier Statement

The realizability barrier leaves a more precise frontier question. If quotient shape alone is too expressive, what should replace it as the organizing unit of the metatheorem?

At minimum, the tractability frontier should not classify arbitrary sets of decision problems. It should range over representation-level structural classes and it should ignore benign changes of presentation.

The right eventual condition list will almost certainly be stronger than this minimal principle. For example, a mature frontier theorem may also require closure under adding explicitly irrelevant coordinates, adjoining dummy actions, or other representation-preserving refinements. But action/state relabel invariance is the irreducible starting point.

Proposition 6.1³⁸ (Exact Certification Depends Only on the Decision Quotient Relation). *If two decision problems on the same state space induce the same decision-equivalence relation, then they induce the same exact-certification structure. In particular, they have the same sufficient coordinate sets, the same minimal sufficient coordinate sets, and the same relevant and irrelevant coordinates. Moreover, their quotient objects are canonically equivalent; in finite settings, those quotient objects therefore have the same cardinality.*

Proposition 6.1 is the canonical statement behind the frontier program. Exact certification is determined by the decision quotient relation itself. Equality of optimizer maps is only a sufficient route to this conclusion, because equal optimizer maps induce the same decision-equivalence relation.

Corollary 6.2³⁹ (Sufficiency Is Relation Refinement). *A coordinate set I is sufficient if and only if the coordinate-agreement relation induced by I refines the decision quotient relation.*

Corollary 6.2 is the cleanest relational form of exact certification. It removes utility language entirely: sufficiency asks whether one state relation is contained in another.

Corollary 6.3⁴⁰ (Relevance Is Failure of Erased Sufficiency). *A coordinate i is irrelevant if and only if deleting it leaves a sufficient coordinate set, namely $\text{univ} \setminus \{i\}$. Equivalently, i is relevant if and only if that erased set is not sufficient.*

³⁷ : FR7–8 ³⁸ : FR30–35, FR37, FR51 ³⁹ : FR48 ⁴⁰ : FR49–50

Corollary 6.3 makes the role of individual coordinates canonical: relevance is exactly the obstruction to retaining sufficiency after coordinate erasure.

Corollary 6.4⁴¹ (Certification Statistics Factor Through the Quotient Relation). *Any statistic that is a function of the sufficient-set family and relevant-coordinate set is invariant under equality of the decision quotient relation. In particular, the number of sufficient coordinate sets, the number of minimal sufficient coordinate sets, and the number of relevant coordinates are quotient invariants.*

Corollary 6.4 records the next logical consequence of Proposition 6.1. Once exact certification is identified with quotient data, every certification statistic built from sufficient sets or relevant coordinates factors automatically through that quotient data as well.

Proposition 6.5⁴² (Zero-Distortion Summaries Refine the Optimizer Quotient). *Let $\sigma : S \rightarrow C$ be a summary map. Suppose that*

$$\sigma(s) = \sigma(s') \implies \text{Opt}(s) = \text{Opt}(s') \quad \text{for all } s, s' \in S.$$

Then each summary fiber is contained in a single optimizer-quotient class. Equivalently, σ factors through the optimizer quotient relation, and distinct optimizer classes require distinct summary symbols.

Proof. If $\sigma(s) = \sigma(s')$, the hypothesis gives $\text{Opt}(s) = \text{Opt}(s')$. So every σ -fiber is contained in one decision-equivalence class. This is exactly the statement that σ refines the optimizer quotient. Conversely, if two distinct optimizer classes shared a summary symbol, choosing representatives from those classes would violate the zero-distortion premise. ■

Proposition 6.5 explains why exact relevance is not merely one point on an approximation spectrum. Once distortion is required to be zero, lossless decision-preserving abstraction is already constrained by the optimizer quotient itself. Later propositions in this section sharpen the approximation boundary: approximate surrogate claims about relevance need explicit stability control relative to the exact optimizer sets, because uniform closeness alone does not preserve the exact decision boundary.

Corollary 6.6⁴³ (Constant-Optimizer Collapse Forces Trivial Certification). *If the optimizer set is constant across all states, then the empty coordinate set is sufficient and every coordinate is irrelevant.*

Corollary 6.6 isolates the first genuinely degenerate mechanism in the tractable basis. The certification problem disappears because there is no state dependence left to certify.

Proposition 6.7⁴⁴ (Explicit Enumeration Is Parameter-Dependent, Not Structural). *For every fixed exponent k , there exists a Boolean-cube family with state space size $2^n > n^k$. Thus explicit-state enumeration can outrun any fixed monomial bound in the ambient dimension parameter.*

Proposition 6.7 is weaker than a full polynomial lower-bound metatheorem, but it already captures the structural point needed here: tractability by brute-force state enumeration depends on a separate size parameter and does not arise from an optimizer-compatible mechanism like symmetry, low rank, or tree structure.

Proposition 6.8⁴⁵ (Positive Affine Utility Reparameterizations Are Invisible). *Statewise positive affine transformations of utility leave exact certification unchanged. In particular, replacing*

$$U(a, s) \quad \text{by} \quad \alpha(s) + \beta(s)U(a, s)$$

with $\beta(s) > 0$ for every state s does not change sufficient coordinate sets or relevance judgments.

⁴¹ : FR41–44 ⁴² : FR176–177 ⁴³ : FR52–53 ⁴⁴ : FR66–67 ⁴⁵ : FR13–16

Proposition 6.8 is another direct consequence of Proposition 6.1. It records a standard utility-theoretic invariance in the exact-certification language: only the induced decision quotient matters, not the particular positive affine scale used to encode utility.

Proposition 6.9⁴⁶ (Duplicate Actions and Duplicate States Preserve Certification). *Duplicating a single action without changing its utility profile preserves the decision quotient relation and hence preserves sufficiency and relevance. At the optimizer-set level, the only possible change is the addition of the duplicate action itself: original optimal actions remain optimal, and the duplicate is optimal exactly when the original action was. Duplicating a single state without changing its utility profile preserves the decision quotient up to the obvious quotient equivalence and also preserves sufficiency and relevance.*

Proposition 6.9 gives two more theorem-forced closure laws. These are not modeling conventions; they are consequences of the fact that exact certification factors through quotient data rather than through accidental multiplicity in the presentation.

Proposition 6.10⁴⁷ (Relabeling Invariance Is Forced by Exact Certification). *Action relabeling preserves optimizer equivalence and exact sufficiency. State relabeling does so as well once the coordinate structure is transported along the state bijection. Consequently, any structural tractability frontier for exact relevance certification must be closed under these relabelings.*

Proposition 6.10 is now best read as a corollary of Proposition 6.1. Bijective relabelings do not change the optimizer map up to the obvious identification, so they cannot change exact certification.

The second necessary ingredient is a genuine expressivity restriction. Call a class *kernel-universal* if, for every labeling map $\phi : S \rightarrow T$, it contains some decision problem whose optimizer quotient realizes exactly the kernel partition of ϕ . By Theorem 5.1, kernel universality is an extremely strong property.

The canonical finite invariant here is the size of the optimizer quotient, equivalently the number of distinct optimal-action sets. The quotient is identified with the range of the optimizer map, so quotient size is not auxiliary bookkeeping; it is intrinsic.

Proposition 6.11⁴⁸ (Quotient Size Is Unbounded Under Realizability). *For every $m \in \mathbb{N}$, there exists a finite decision problem whose optimizer quotient has size exactly m .*

Proof. Take $m = k + 1$ and realize the identity labeling on $\text{Fin}(k + 1)$. Proposition 5.3 identifies the quotient object with the range of that identity labeling, which has cardinality $k + 1$. Equivalently, Proposition 5.2 realizes the discrete equivalence relation on $\text{Fin}(k + 1)$. ■

A meaningful frontier statement cannot range over classes that are simultaneously relabel-invariant and kernel-universal, because quotient shape alone then becomes too expressive to isolate the boundary.

Proposition 6.12⁴⁹ (Invariance Alone Is Too Weak). *The universal class of all decision problems satisfies action relabeling invariance and state relabeling invariance, but it is kernel-universal. Consequently, relabeling invariance by itself is far too weak to isolate a tractability frontier.*

Proposition 6.12 separates two kinds of conditions that can otherwise be conflated.

1. **Benign invariance axioms** say the family should ignore arbitrary naming choices.
2. **Expressivity-limiting axioms** say the family should not realize arbitrary quotient structure wholesale.

⁴⁶ : FR54–56, FR68–73 ⁴⁷ : FR17–20 ⁴⁸ : FR24, FR38 ⁴⁹ : FR21–23

Any genuine optimizer-compatible frontier theorem will need both. Without decision-quotient invariance and its immediate corollaries such as relabel invariance and positive affine invariance, the statement is not structural. Without an expressivity restriction excluding kernel universality or something comparably strong, the realizability theorem and Proposition 6.11 make the class too large for quotient shape alone to carry the boundary.

Proposition 6.13⁵⁰ (Independent Summary Frameworks Converge on the Same Structural Core). *Write $r = \text{srnk}(\mathcal{D})$ for the number of relevant coordinates of a Boolean decision problem, and let m be the number of optimizer-quotient classes. Then:*

1. $m \leq 2^r$, so the counting entropy $\log_2 m$ is at most r ;
2. the diagonal 0/1 relevance-support matrix has rank exactly r ;
3. every zero-distortion decision-preserving summary requires at least m distinct summary symbols.

Hence quotient entropy, Fisher-style support counting, and zero-distortion summary size are all controlled by the same support/quotient core.

Proof. If two states agree on all relevant coordinates, then they differ only on irrelevant ones. Changing those irrelevant coordinates one at a time cannot change the optimizer, so the relevant coordinates already determine the optimizer class. Therefore there are at most 2^r optimizer classes, one for each Boolean assignment on the relevant support, and $\log_2 m \leq r$ follows.

For the second clause, the diagonal relevance-support matrix has a 1 exactly on relevant coordinates and a 0 elsewhere. Its rank is therefore the number of relevant coordinates, namely r .

The third clause is exactly Proposition 6.5: a zero-distortion summary must assign distinct symbols to distinct optimizer classes. ■

Proposition 6.13 does not identify entropy, support counting, and lossless summaries as identical notions. It shows that they already converge on the same structural content before any admissibility axiom is imposed. Closure-law invariance follows from this convergence: once several independent frameworks ignore surface presentation in favor of the support/quotient core, a structural tractability classifier should do the same. The remaining admissibility clauses serve as algorithmic and semantic guardrails.

From Necessary Conditions to an Admissibility Class

The closure properties above are theorem-forced by exact-certification semantics, but they are not enough on their own to support a meaningful impossibility theorem. If one allows unrestricted predicates, a semantic predicate can simply encode the target complexity class and trivialize the statement.

Proposition 6.14⁵¹ (Unrestricted Predicates Trivialize Exact Characterization). *Without admissibility restrictions, there exist normalization predicates whose membership condition directly encodes the intended polynomial-time side of the landscape, so unrestricted collapse-impossibility claims fail.*

Proposition 6.14 is not a by-product of the proof method. It is the expected behavior of unrestricted semantic predicates. Therefore any meaningful frontier theorem must be parameterized by an admissibility class that simultaneously (i) respects theorem-forced invariances and (ii) blocks direct semantic encoding.

⁵⁰ : FR178–182 ⁵¹ : FR145

Proposition 6.15⁵² (Closure Operations Preserve Exact Certification). *The theorem-forced closure operations preserve the exact-certification problem itself. More precisely:*

1. *action relabeling, coordinate relabeling, and statewise positive affine reparameterization preserve sufficient coordinate sets and relevant coordinates under the evident transport;*
2. *action duplication and state duplication preserve sufficient coordinate sets and relevant coordinates exactly;*
3. *binary irrelevant-coordinate extension preserves sufficiency after lifting coordinate sets, preserves relevance on the original coordinates, and makes the new coordinate irrelevant.*

In particular, each closure step induces the same exact-certification decision problem up to an explicit coordinate-set transport. For the irrelevant-coordinate case in the present binary-pairwise formalization, the encoding blowup is only constant-factor because the added coordinate is binary.

Proof sketch. Each clause is verified directly in the artifact at the level of exact-certification semantics. The relabeling and positive-affine cases preserve the decision-equivalence relation, hence preserve sufficiency and relevance. The duplication cases preserve decision-equivalence through the explicit quotient projections back to the original problem. The irrelevant-coordinate case is formalized as binary noise extension: $I \subseteq [d]$ is sufficient for the base slice if and only if its lift is sufficient after extension, the original coordinates remain exactly the relevant ones, and the new coordinate is irrelevant. Table 1 summarizes the theorem-backed transport statement. The encoding remarks in the right-hand column are direct from the concrete binary-pairwise representation and are not separate Lean claims. ■

Operation	Exact-certification transport	Encoding effect
Action/state relabeling	same sufficient sets and relevant coordinates after transport	relabeling only
Positive affine reparameterization	same sufficient sets and relevant coordinates	same arity, same action set; utility magnitudes rescaled
Action/state duplication	same sufficient sets and relevant coordinates	carrier duplication only
Binary irrelevant-coordinate extension	$I \leftrightarrow \text{lift}(I)$, old relevance preserved, new coordinate irrelevant	arity increases by one binary coordinate

Table 1: Closure-operation transport for exact certification. The transport column is theorem-backed in Lean; the encoding-effect column records the direct representation-level effect of each operation in the binary-pairwise formalization.

Definition 6.16 (Admissible Normalization Predicate). A normalization predicate is *admissible* when it satisfies all of the following:

1. polynomial-time checkability at the slice level;

⁵² : FR15-20, FR55-60, FR83-85

2. invariance under the closure laws forced by exact certification (relabelings, positive affine reparameterizations, duplication operations, and binary irrelevant-coordinate extension);
3. structural extractability of the associated dependency graph;
4. bounded-pattern definability.

More explicitly, let $|U|$ denote the encoding size of a binary pairwise slice U , and let $X(U)$ be its canonical finite pairwise syntax. Polynomial-time checkability means that there exist constants $c, k \in \mathbb{N}$ and a decision procedure A such that

$$A(U) = 1 \iff Q(U), \quad \text{time}_A(U) \leq c(|U| + 1)^k + c.$$

Structural extractability means that whenever $Q(U)$ holds, the associated coordinate graph is obtained from syntax alone: there exists a uniform extractor

$$E : X \mapsto G_E(X)$$

on finite pairwise syntactic presentations such that the graph attached to U is exactly $G_E(X(U))$.

Bounded-pattern definability is likewise finite and explicit. There must exist uniform bounds on radius, neighborhood size, action alphabet size, and coefficient magnitude, together with finite lists of permitted and forbidden rooted local patterns, such that membership in Q is determined by the presence of one witness pattern or the absence of all forbidden patterns in the rooted interaction neighborhoods of $X(U)$. No unbounded global computation is hidden inside the definition.

Polynomial-time checkability is the minimal algorithmic requirement. A tractability characterization that cannot itself be recognized efficiently has little explanatory value: it would only relocate the computational difficulty from exact certification to the membership test for the classifier.

Closure-law invariance is the semantic requirement used in the orbit-witness arguments of Section 4. It is forced by the Block 6 invariance theorems, because exact certification itself is unchanged by relabelings, positive affine utility reparameterizations, duplication operations, and binary irrelevant-coordinate extensions.

Structural extractability distinguishes structural characterizations from purely semantic ones. Without such a condition, one can define predicates directly on optimizer behavior or on the solved certification instance itself, bypassing the stated aim of classifying tractability by representation-level structure.

Bounded-pattern definability is the locality guardrail. Its role is to exclude predicates whose membership test secretly performs an unbounded global computation under a structural name. In particular, it prevents one from reintroducing oracle power through an arbitrarily large finite schema.

Theorem 6.17 (Tractability Classifiers Are Forced to Be Closure-Invariant). *Let C be any procedure assigning to each representation in a closure-closed domain Γ a verdict in $\{\text{tractable}, \text{intractable}\}$, and suppose that C correctly predicts whether exact certification is polynomial-time decidable on the underlying problem. Then C must agree on representations within the same closure orbit. Equivalently, every correct tractability classifier on a closure-closed domain is closure-invariant on that domain, regardless of whether its internal features are themselves invariant.*

Proof sketch. This is the tractability instance of the abstract correctness-on-domain theorem formalized in Lean. If $U, V \in \Gamma$ lie in the same closure orbit, Proposition 6.15 shows that they induce the same exact-certification problem up to the explicit coordinate-set transport attached to the relevant closure step. The theorem uses the semantic notion of correctness stated above: tractability

is attached to that underlying exact-certification problem, not to arbitrary relabelings or duplicate-label presentations of the same problem. Correctness of C therefore forces the same verdict on both. ■

Theorem 6.18⁵³ (Compute-Cost Version: Optimizer Computation). *Let C be any procedure assigning to each representation in a closure-closed domain Γ a verdict in $\{\text{tractable}, \text{intractable}\}$, and suppose that C correctly predicts whether optimizer computation is polynomial-time solvable on the underlying problem. Then C must agree on representations within the same closure orbit.*

Proof sketch. This is the optimizer-computation instance of the same correctness-on-domain theorem. The compute-cost transport layer in the artifact shows that each closure step preserves polynomial-time solvability of optimizer computation under its explicit state and output transports. Correctness of C therefore forces the same verdict on closure-equivalent representations. ■

Corollary 6.19⁵⁴ (Compute-Cost Version: Canonical Payload and Search Tasks). *The same closure-orbit agreement conclusion holds when C correctly predicts polynomial-time solvability of deterministic optimizer-set payload output. It also holds when C correctly predicts polynomial-time solvability of admissible-output search on optimizer-set semantics.*

Proof sketch. Apply the same correctness-on-domain theorem to the two canonical compute-cost families built from optimizer-set payload output and optimizer-set admissible-output search. ■

Thus the forced-invariance argument applies not only to exact-certification predicates, but also to all three canonical output-production tasks treated in the compute-cost layer.⁵⁵

This answers the natural objection that Section 4 excludes only classifiers that advertise closure invariance as an explicit axiom. Correctness already forces the same orbit agreement. The closure laws therefore supply the representation-level congruence needed for the orbit-gap program: once two slices are related by these theorem-forced presentation equivalences, any correct tractability classifier must identify them. Theorem 4.9 therefore applies to any correct classifier on a closure-closed domain once the target notion is tractability itself: if the classifier separates same-orbit witnesses, it is simply wrong on at least one of them.

Proposition 6.20⁵⁶ (The Admissibility Layer Is Nonempty). *Definition 6.16 is not vacuous. There exist admissible normalization predicates; in fact both the constant-true and constant-false slice predicates are admissible.*

Proof sketch. Both predicates are decidable in constant time and are automatically invariant under all closure laws. Their graph extractors are fixed graphs independent of the input slice. For bounded-pattern definability, use a single impossible local pattern: a radius-zero pattern with two distinct vertices cannot occur in any slice. Taking that pattern as forbidden yields the constant-true predicate, and taking it as a required witness yields the constant-false predicate. ■

Thus the impossibility theorem is not an emptiness statement about Definition 6.16. The admissibility layer already contains explicit predicates; the point of the no-go theorem is that correctness across the obstruction families forces a contradiction once closure-orbit agreement is respected.

Proposition 6.21⁵⁷ (Bounded Distinct Action Profiles Compress to Bounded Actions). *For a binary pairwise slice U , let $d(U)$ be the number of distinct action utility profiles. Then there exists a compressed slice U^{prof} with exactly $d(U)$ actions such that a coordinate set is sufficient for U if and*

⁵³ : FR197–199, FR245–249 ⁵⁴ : FR248 ⁵⁵ : FR245–249, FR259–262 ⁵⁶ : FR200 ⁵⁷ : FR205–209

only if it is sufficient for U^{prof} , and a coordinate is relevant for U if and only if it is relevant for U^{prof} . Consequently, if $d(U) \leq k$, the bounded-actions tractability theorem applies to U^{prof} , so the bounded-distinct-profile subcase is polynomial on the exact-certification side.

Proof sketch. Replace each action by its full utility profile over states and quotient the action set by profile equality. Duplicated actions collapse to a single profile action, but exact certification is unchanged because optimizer equality depends only on which profiles are optimal, not on how many labels realize a given profile. The compressed slice has exactly $d(U)$ actions by construction, and the artifact proves equality of the sufficient-coordinate and relevant-coordinate predicates between U and U^{prof} . The existing bounded-actions polynomial-time theorem then applies to the compressed slice. ■

Proposition 6.21 isolates a genuine positive tractable subcase by compression to bounded actions. It is stated as a compression theorem rather than direct admissibility membership because Definition 6.16 fixes a bounded-pattern action alphabet in advance. The proposition still shows that nontrivial positive classification work is available through closure-law-respecting reduction inside the present framework.

Proposition 6.22⁵⁸ (Bounded-Pattern Predicates Stabilize Above a Finite Action Bound). *For every bounded-pattern definable predicate Q , there exists a finite bound B such that Q is constant on all slices with more than B actions. Equivalently, once the action alphabet exceeds the action bound built into the defining pattern scheme, the scheme can no longer distinguish between such slices.*

Proof sketch. In a bounded-pattern scheme every witness and forbidden pattern has action alphabet size at most a fixed bound B . If a slice has more than B actions, then no listed local pattern can occur in it, because occurrence requires an action-set equivalence between the pattern and the slice. The witness branch of the scheme is therefore impossible, while the forbidden branch is determined only by whether the forbidden list is empty. Hence the predicate is constant above the fixed action bound. ■

This is why Proposition 6.21 is stated as a compression theorem rather than as direct admissibility membership for the distinct-profile predicate. Unbounded profile counting does not naturally fit a definition whose bounded-pattern clause fixes a finite action alphabet in advance.

Proposition 6.23⁵⁹ (Deterministic Payload Transfer). *Let S be a coordinate space, let $\phi : S \rightarrow T$ be any deterministic payload map into a finite label type, and let D_ϕ be the induced decision problem with action space T and utility*

$$U(a, s) = \begin{cases} 1, & a = \phi(s), \\ 0, & a \neq \phi(s). \end{cases}$$

Then every coordinate set is sufficient for ϕ if and only if it is sufficient for D_ϕ , and every coordinate is relevant for ϕ if and only if it is relevant for D_ϕ . Equivalently, exact feature sufficiency and relevance for any deterministic payload reduce definitionally to exact relevance certification for an induced decision problem.

Proof sketch. For the induced decision problem, the optimizer at state s is the singleton $\{\phi(s)\}$. Hence two states have the same optimizer if and only if they carry the same payload value. The sufficiency and relevance predicates are therefore literally the same coordinate conditions on S in both formulations. ■

⁵⁸ : FR214–216 ⁵⁹ : FR217–222

This is a theorem of exact semantic equivalence. Deterministic payload sufficiency and relevance are already exact relevance certification for the induced decision problem.

Corollary 6.24⁶⁰ (Nonempty Set-Valued Payload Transfer). *Let $F(s) \subseteq A$ be a nonempty feasible-action set at each state of a coordinate space, and equip allowed actions with utility u_{allowed} and blocked actions with utility u_{blocked} where $u_{\text{blocked}} < u_{\text{allowed}}$. Then exact feature sufficiency and relevance for the set-valued payload F are equivalent to exact relevance certification for the induced decision problem whose optimal set at state s is exactly $F(s)$.*

Proof sketch. This is the set-valued analogue of Proposition 6.23. The strict utility gap makes the optimizer set coincide with the feasible-action set at every state, so the same coordinate conditions define sufficiency and relevance on both sides. ■

Corollary 6.24 covers total search semantics stated as nonempty feasible-witness sets. Deterministic payloads are the singleton special case.

Corollary 6.25⁶¹ (Arbitrary Set-Valued Payload Transfer via Failure Tokens). *Let $F(s) \subseteq A$ be any set-valued payload, possibly empty. Form the totalized payload on $\text{Option}(A)$ by adjoining a distinguished failure token whenever $F(s) = \emptyset$. Then exact feature sufficiency and relevance for F are equivalent to exact relevance certification for the induced decision problem on the totalized payload.*

Proof sketch. Map each admissible output $a \in F(s)$ to $\text{some}(a)$, and use none exactly when $F(s)$ is empty. Equality of the original set-valued payloads is then equivalent to equality of the totalized nonempty payloads, so Corollary 6.24 applies after this totalization step. ■

Corollary 6.25 removes the empty-fiber caveat for search-style semantics. Set-valued payloads with failure states transfer without choosing a deterministic witness selector.

Corollary 6.26⁶² (Arbitrary Exact Output Semantics Transfer). *Let $R(s, a)$ be any state-indexed admissible-output relation. Then exact feature sufficiency and relevance for the induced output semantics reduce to exact relevance certification for a decision problem obtained by totalizing the admissible-output sets with a failure token and assigning a strict allowed-versus-blocked utility gap.*

Proof sketch. Write $F(s) = \{a : R(s, a)\}$. This is exactly the arbitrary set-valued case of Corollary 6.25. The relation form is only a notational repackaging of the same theorem. ■

Any output specification is in scope once written as a state-indexed admissible-output relation. This includes decision, counting, search, approximation guarantees, PAC/regret/risk guarantees, anytime or finite-horizon guarantees, distributional specifications, and any other specification stated by the outputs that count as correct at each state.

Corollary 6.27⁶³ (Statistical Guarantee Semantics Transfer). *Let $R(s, a)$ specify the admissible outputs for a PAC guarantee, regret guarantee, statistical risk guarantee, anytime guarantee, or finite-horizon guarantee at state s . Then exact feature sufficiency and relevance for that guarantee semantics reduce to exact relevance certification for the induced decision problem.*

Proof sketch. Each case is a named specialization of Corollary 6.26. The output object may be a hypothesis, learner, estimator, or policy; correctness still means exact agreement with the admissible-output relation specifying the guarantee. ■

⁶⁰ : FR223–225 ⁶¹ : FR226–228 ⁶² : FR229–231 ⁶³ : FR268–282

Corollary 6.28 (Statistical Guarantee Semantics Inherit the Closure-Orbit Consequences). *Once a PAC, regret, statistical-risk, anytime, or finite-horizon guarantee is written as a state-indexed admissible-output relation, the same closure-orbit agreement and no-go theorems apply to classifiers for that semantics. In particular, Theorem 6.17, Corollary 4.11, and Corollary 4.14 apply unchanged after the transfer of Corollary 6.27.*

Proof sketch. Corollary 6.27 reduces each named statistical guarantee semantics to exact relevance certification for the induced decision problem. The closure-orbit agreement and no-go theorems are already proved for exact relevance certification itself. Statistical guarantee semantics also inherit the same exact-semantics quotient invariance, because their sufficient-coordinate family and relevant-coordinate set depend only on admissible-output equivalence. ■

Corollary 6.29⁶⁴ (Exact Approximation Semantics Transfer). *Let $R(s, a)$ specify the admissible outputs for an exact approximation specification at state s , for example the set of all outputs meeting a fixed approximation guarantee. Then exact feature sufficiency and relevance for that approximation semantics reduce to exact relevance certification for the induced decision problem.*

Proof sketch. This is the approximation-specialized form of Corollary 6.26. Approximation enters only through the specification: once the admissible-output relation says which outputs meet the guarantee, correctness with respect to that approximation specification is exact agreement with that relation. ■

Proposition 6.30⁶⁵ (Approximate Relevance and Sufficiency Claims Need Explicit Stability Control). *Let D and D' be decision problems on the same coordinate space, and suppose their utilities are uniformly δ -close. If coordinate i has a relevance witness in D given by states s, s' with distinct strict optimal actions at the two witness states, and if the strict utility gap at each witness state exceeds 2δ , then i is also relevant in D' . Likewise, if a coordinate set I has a non-sufficiency witness in D given by states s, s' that agree on I but have distinct strict optimal actions, and if the strict utility gap at each witness state exceeds 2δ , then I is still not sufficient in D' .*

Proof sketch. The existing uniform-approximation theorem preserves the optimizer set at any state whose strict utility gap dominates the approximation error. Applying that theorem to both witness states keeps the two singleton optimizer sets distinct. In the relevance case this preserves the same relevance witness. In the sufficiency case the same state pair still agrees on I but still has different optimizer sets, so it remains a non-sufficiency witness. ■

Corollary 6.31⁶⁶ (Arbitrarily Small Uniform Perturbations Can Flip Relevance). *For every $\varepsilon > 0$, there exist two decision problems on the same one-coordinate Boolean state space that are uniformly ε -close, yet the unique coordinate is relevant in one problem and irrelevant in the other.*

Proof sketch. The artifact contains an explicit two-action, one-coordinate construction. In the first problem the optimizer tracks the Boolean state, so the unique coordinate is relevant. In the second problem all actions are tied at every state, so the coordinate is irrelevant. The two utility functions differ uniformly by at most ε . ■

Corollary 6.32⁶⁷ (Arbitrarily Small Uniform Perturbations Can Flip Sufficiency). *For every $\varepsilon > 0$, there exist two decision problems on the same one-coordinate Boolean state space that are uniformly ε -close, yet the empty coordinate set is sufficient in one problem and not sufficient in the other.*

⁶⁴ : FR232–234 ⁶⁵ : FR263, FR266 ⁶⁶ : FR264 ⁶⁷ : FR267

Proof sketch. The same one-coordinate construction separates the constant-optimizer case from the state-tracking case. In the flat problem the empty set is sufficient because the optimizer set is constant, while in the state-tracking problem the empty set is not sufficient because the two Boolean states induce different optimizer sets. The two utility functions are still uniformly ε -close. ■

Thus closeness alone does not justify an approximate relevance or sufficiency claim. Any ε -based surrogate claim needs an explicit reduction to the exact optimizer sets, for example by a witness-gap bound as in Proposition 6.30, because otherwise arbitrarily small perturbations can change the exact judgment.

Proposition 6.33⁶⁸ (Exact Semantic Claims Depend Only on Admissible-Output Equivalence). *Let R and R' be two exact output semantics on the same state space. If they induce the same equality relation on admissible-output sets, then they induce the same sufficient coordinate sets and the same relevant coordinates. Exact semantic claims about sufficiency and relevance therefore depend only on the induced admissible-output equivalence relation.*

Proof sketch. The transfer theorems reduce both semantics to exact relevance certification for induced decision problems. If the admissible-output equality relation is the same, the induced sufficiency and relevance structures are the same as well. The artifact proves equality of the sufficient-set family and of the relevant-coordinate set under this hypothesis. ■

Corollary 6.34⁶⁹ (Every State Equivalence Relation Is Realizable as Exact Output Semantics). *For every equivalence relation on the state space, there exists an exact output semantics whose admissible-output equality relation is exactly that equivalence relation.*

Proof sketch. Take the output space to be the quotient by the given equivalence relation and assign to each state its own quotient class as a singleton admissible-output set. Equality of those singleton output sets is then exactly the original equivalence relation. ■

Proposition 6.35⁷⁰ (Semantically Extensional Claims Factor Through the Exact-Semantics Quotient). *Let C be any claim about exact output semantics that depends only on admissible-output equivalence. Then C factors through the quotient of output semantics by admissible-output equivalence. In particular, once a semantic claim is exact and extensional, the exact-semantics quotient is the object of the claim.*

Proof sketch. This is the quotient-factorization theorem for semantically extensional maps, specialized to **Prop**-valued claims. If C takes the same truth value on any two semantics with the same admissible-output equivalence, then C descends to the quotient by that equivalence relation. ■

Closure-law invariance (clause 2) is not an arbitrary stylistic restriction. Theorem 6.17 shows that correctness already forces closure-orbit agreement, and Theorem 4.9 shows that this theorem-forced congruence is the only load-bearing hypothesis in the no-go theorem. The remaining clauses are the minimal guardrails against the failure modes identified in Proposition 6.14; the specific guardrail formulation of clauses 1, 3, and 4 is one natural choice but is not load-bearing.

⁶⁸ : FR235–236 ⁶⁹ : FR237, FR240 ⁷⁰ : FR241–244

7 Related Work

Rough Sets, Feature Selection, and Exact Relevance

At the static level, rough-set reduct theory [19, 30] is the clearest classical comparison point. That literature studies which attributes can be deleted while preserving decision distinctions. Exact static sufficiency can be viewed as a reduct condition for the induced decision table. The question here is different: which tractable mechanisms admit a finite characterization?

A separate literature studies feature selection, variable importance, and model explanation in machine learning and AI, including wrapper and filter methods [15], general surveys [13], and attribution methods based on Shapley-style decompositions [16]. These works matter here by motivation rather than by technique. They typically optimize predictive performance, explanatory salience, or approximation quality under data-dependent criteria. Exact relevance certification instead asks for a semantic preservation property: which coordinates are necessary to preserve the outputs that a specification counts as correct. Section 6 proves the transfer to exact relevance certification for arbitrary state-indexed admissible-output specifications. This includes decision, counting, search, approximation guarantees, and distributional specifications once written as admissible-output relations.

Decision-theoretic informativeness and value of information, beginning with Blackwell’s comparison of experiments and subsequent decision-theoretic treatments [4, 23, 14], provide a second comparison point. Both settings ask when one representation retains all decision-relevant distinctions of another. The difference here is the combinatorial complexity of certifying exact preservation under coordinate deletion.

Information-Theoretic and Transport Viewpoints

The zero-distortion discussion belongs to the classical information-theoretic tradition of exact distinguishability and lossless coding boundaries, going back to Shannon and standard modern treatments of source coding and rate-distortion theory [25, 26, 27, 6]. Fisher-style support counting supplies another classical comparison lens on relevant support [8]. These viewpoints are used only structurally. There is no general coding or statistical estimation problem here. Zero distortion, entropy, and support counting appear only to show that several independent summary frameworks are already constrained by the same optimizer-quotient core. The Wasserstein language in Section 4 plays the same role: a qualitative description of quotient branching rather than a full transport-theoretic model.

Backdoors, Tractable Islands, and Dichotomy Programs

The nearest complexity-theoretic analogue is backdoor tractability for constraint satisfaction [31, 3, 9]. There, a small structural set exposes membership in a tractable class even when the ambient problem is hard. The comparison is direct: exact relevance certification also asks which coordinates matter, and polynomial-time behavior emerges when the source of hardness is restricted by structure. Bessiere et al. [3] make especially clear that exploiting a known tractable structure and discovering the responsible structure are different algorithmic tasks; later parameterized work sharpened that distinction. This is closely aligned with the present paper’s separation between positive tractable mechanisms and the harder meta-level problem of recognizing a correct frontier classifier.

The broader analogy is the dichotomy tradition for satisfiability and constraint satisfaction, from Schaefer’s Boolean dichotomy theorem [24] to the finite-domain CSP dichotomy theorems of Bulatov and Zhuk [5, 34]. The expectation that a clean tractability boundary should exist was

articulated in the Feder–Vardi program [7], and later algebraic work of Barto and Kozik clarified why finite structural boundaries are plausible in that setting [2]. The point here is narrower: for output specifications of any kind, quotient realizability and closure-law invariance defeat the most direct admissible classifier, both at the feature-sufficiency layer and at the compute-cost layer of optimizer production. This is compatible with the CSP dichotomy program and helps explain its form: once admissible closure-invariant classifiers are ruled out, positive frontier theorems must use stronger algebraic structure outside Definition 6.16. The open problem is to identify an analogue of that stronger structure for the present setting.

The quotient viewpoint also touches the literature on exact abstraction and aggregation in stochastic control, Markov decision processes, and reinforcement learning [18, 11, 20]. Those works study when states may be aggregated while preserving value or policy structure. Our setting is narrower: we ask for exact preservation of optimizer classes under coordinate-hiding maps rather than arbitrary state abstractions. Still, the common theme is that semantic invariants of decision behavior should survive presentation-level simplification.

Meta-Impossibility Traditions

The closest analogy is Rice-style impossibility. Rice’s theorem [21] shows that nontrivial extensional properties of partial recursive functions are undecidable, with extensionality forced by what “semantic” means rather than chosen as an axiom. Refinements such as the Myhill–Shepherdson theorem and the Rice–Shapiro theorem extend this pattern to broader structured semantic domains [17, 28]. The present theorem has the same structure with a different forced invariance. Proposition 6.1 shows that exact certification depends only on quotient data, and Theorem 6.17 lifts this to tractability classification: any correct tractability classifier, for any output specification written as an admissible-output relation, must agree on closure-equivalent representations. The no-go then depends on this forced invariance alone (Theorem 4.9), with Corollary 4.11 giving the resulting impossibility for correct classifiers. The remaining admissibility clauses play the role of algorithmic and structural guardrails that exclude classifiers admissible in name only, but they do not enter the contradiction.

At the foundational level, Gödel’s incompleteness theorems show that sufficiently expressive formal systems cannot internalize all truths about their own proof theory [12]. In learning and optimization, no-free-lunch results show that universal guarantees disappear on overly broad problem domains [32, 33]. In social choice, Arrow’s theorem and the Gibbard–Satterthwaite theorems show that natural axiomatic requirements become jointly unsatisfiable on rich enough domains [1, 10, 22]. We place the present theorem in that same impossibility tradition: a natural invariance package is fixed, and no correct classifier survives on the intended domain.

8 Conclusion

The frontier question now has both a positive and a negative answer. On the positive side, the tractable landscape studied here factors into primitive mechanisms, lifts, and degeneracies. Proposition 6.21 shows that this positive side remains substantive at the representation level: bounded distinct action profiles compress to bounded-action slices without changing exact certification, so the existing bounded-actions theorem already yields a genuine structural classifier on that subcase. Section 6 shows that the framework is not confined to native decision outputs. It captures the structural-sufficiency problem for every output specification expressible as a state-indexed admissible-output relation. This includes decision, counting, search, approximation guarantees, and

distributional specifications once written in that form. Proposition 6.33, Corollary 6.34, and Proposition 6.35 sharpen that point: semantic claims depend only on admissible-output equivalence, every state equivalence relation is already realizable at the output-semantics layer, and semantically extensional claims factor through that quotient. The Lean development also formalizes the compute-cost closure-orbit layer: optimizer computation, deterministic optimizer-set payload output, and admissible-output search each come with canonical closure-law transport, and correctness of a polynomial-time classifier again forces closure-orbit agreement. Taken together, these results constrain tractability classification at the level of computational specifications rather than at the level of any one output formalism. On the negative side, quotient realizability is maximal, so quotient shape cannot classify tractability by itself, and an admissibility-layer collapse theorem rules out direct finite classifiers even after imposing theorem-forced closure laws together with computational and structural restrictions. Moreover, Theorem 4.9 isolates the load-bearing hypothesis: the collapse depends only on closure-law invariance, so any guardrail package whose admissible predicates respect this single forced invariance inherits the impossibility. Theorem 6.17 strengthens the conclusion further: correctness itself forces closure-orbit agreement, and Corollary 4.11 gives the resulting universal-scope no-go for correct classifiers on closure-closed domains.

Technically, the mechanism is closure-orbit disagreement via pair-targeted action-independent affine transport. The same template now applies to four obstruction families (dominant pair, margin masking, ghost action, and additive/statewise offset), yielding an admissible-collapse theorem. The dominant-pair witness also yields a concrete compute-cost no-go: no correct classifier for polynomial-time optimizer computation can decide dominant-pair status exactly, so the compute-cost layer fires on the obstruction families in the same theorem-backed way as the admissible-predicate layer.

The open problem is now sharper. Simple finite lists and unconstrained quotient predicates cannot characterize the boundary, and admissible closure-invariant classifiers of the present kind cannot do so either. Any successful frontier theorem must therefore invoke stronger representation-sensitive structure than the one ruled out here.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used Codex, Claude Code, Augment, Kilo, and OpenCode in order to refine prose, clean up notation, edit the $\text{\LaTeX}$ source, and test alternative phrasings and structural arrangements. After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

Generative AI tools were used throughout the manuscript, across all sections and across all stages from initial drafting to final revision. The tools were used for boilerplate generation, prose and notation refinement, $\text{\LaTeX}$ /structure cleanup, translation of informal proof ideas into candidate formal artifacts (Lean/ $\text{\LaTeX}$), and repeated adversarial reviewer-style critique passes to identify blind spots and clarity gaps.

The author retained full intellectual and editorial control, including problem selection, theorem statements, assumptions, novelty framing, acceptance criteria, and final inclusion/exclusion decisions. No technical claim was accepted solely from AI output. Formal claims reported as machine-verified were admitted only after Lean verification (no `sorry` in cited modules) and direct author review. The author is solely responsible for all statements, citations, and conclusions.

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